

ABSTRACT

The Smart Sea Belt is a proposed intelligent marine safety system designed to improve the safety of fishermen and other small-vessel occupants during emergency situations at sea. Marine environments can expose individuals to risks such as accidental falls overboard, loss of balance, delayed emergency communication, sudden health-related incidents and difficulty in reaching a safe location after an emergency.

The proposed system integrates wearable sensing, wireless communication, emergency flotation and an automatic rudder diversion mechanism into a coordinated safety architecture. Each Smart Watch unit can monitor parameters such as heart rate and motion and provides an SOS interface. Bluetooth Low Energy (BLE) is used for communication between wearable units and a centralized Boat Hub. The Boat Hub combines the wearable data with GPS information and forwards relevant emergency information through LoRa communication to a Base Unit. The Base Unit supports monitoring and emergency alerts.

For an emergency involving a person overboard or a critical safety event, the system is designed to support two physical safety responses: an automatically deployable flotation mechanism and an automatic rudder diversion mechanism. The flotation subsystem is intended to assist the person in maintaining buoyancy, while the rudder mechanism is intended to redirect the boat toward the relevant emergency location or safer recovery direction, subject to the vessel's operating conditions and control logic.

The proposed architecture follows the sequence “Sense → Process → Detect → Communicate → Alert”. The system also provides a foundation for an IoT-enabled mobile application through which boat operators, authorized monitoring personnel and rescue stakeholders can view alerts and status information. The estimated prototype development cost is ₹58,000, covering the base unit, Boat Hub, five wearable units, integration/testing and marine protection.

This project report presents the problem background, objectives, literature-oriented context, proposed architecture, components, working methodology, automatic rudder mechanism, software and communication design, cost structure, feasibility, limitations, future scope and commercialization considerations.